

README

DocProcessor

Documentation processing and feature map extraction for QA automation.

license Apache 2.0 go 1.25+ anti-bluff CONST-035 i18n CONST-046

Overview

DocProcessor is a **standalone, project-not-aware, fully decoupled** Go module (per CONST-051(B)) that loads project documentation, builds structured feature maps, and tracks verification coverage. It is designed to work with LLM agents for intelligent feature extraction, but also includes heuristic-based extraction for offline use.

Round 220 (2026-05-19) deep-doc + test-matrix enrichment. This README matches actual capability — every claim below is exercised by an automated test or Challenge script in this repository (per CONST-048 invariants 1, 3, 5, 6 and the §11.4 anti-bluff covenant).

Verbatim 2026-05-19 operator mandate (CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"*

Quick Start

```
# Clone (per CONST-056: install upstreams immediately if present)
git clone git@github.com:HelixDevelopment/DocProcessor.git
cd DocProcessor
```

```

install_upstreams                                # if Upstreams/ recipe dir
    present

# Build
go build ./...

# Run unit + integration + stress + security + E2E + automation
    tests
go test ./... -race -count=1

# Build the CLI
go build -o bin/docprocessor ./cmd/docprocessor

# Run against a docs directory
./bin/docprocessor /path/to/docs                # terse summary
./bin/docprocessor --verbose /path/to/docs        # per-feature/
    screen/workflow lines

```

Architecture

DocProcessor is organised into seven packages — six domain packages plus pkg/i18n for the CONST-046 string-externalisation contract:

Package	Purpose
pkg/loader	Document loading + parsing (Markdown, YAML, HTML, AsciiDoc, RST)
pkg/feature	Feature extraction, FeatureMap building, FeatureMapBuilder
pkg/coverage	Thread-safe coverage tracking with RWMutex
pkg/docgraph	Inter-document link graph with JSON/Mermaid export
pkg/llm	LLMAgent interface + prompt templates (no hard dependency on any provider)
pkg/config	Configuration loading from .env files
pkg/i18n	Translator contract + NoopTranslator default (CONST-046 no-hardcoded-content)

Processing Pipeline

Load Docs -> Parse Sections -> Extract Features -> Build FeatureMap -> Enrich (LLM) -> Track Coverage

1. **Load & Parse** — scan project tree for documentation files in configured formats.
2. **Extract Features** — heuristic extraction (offline) or LLM-powered extraction.

3. **Build Feature Map** — structured, queryable map with categories + platform matrix.
4. **Enrich** — optional LLMagent infers screens and generates test steps.
5. **Track Coverage** — thread-safe per-platform verification tracking.

CLI surface (round 209 grammar)

docprocessor [--verbose|-v] <docs-directory>

Flag	Effect
--	Emit per-feature / per-screen / per-workflow lines after the summary block.
-v	Alias for --verbose.
(no flag)	Terse summary: format-list, loaded-count, summary-header, feature-map/ doc-graph counts, per-category + per-platform lines, completion line.

Wire-output message IDs (CONST-046)

Every user-facing line is emitted via the injected pkg/i18n.Translator, never as a hardcoded English literal. The full message-ID catalogue:

Round	Message ID	Emitted when
97	docprocessor_cli_usage	argv < 2 — usage line
97	docprocessor_cli_error_loading_docs	loader returns an error
97	docprocessor_cli_loaded_documents	loader returns N documents
97	docprocessor_cli_error_building_feature_map	feature builder returns an error
97	docprocessor_cli_feature_map_summary	summary block — F / S / W counts
97	docprocessor_cli_doc_graph_summary	summary block —

Round	Message ID	Emitted when
		nodes / edges
97	docprocessor_cli_category_line	per-category count line
97	docprocessor_cli_platform_line	per-platform count line
209	docprocessor_cli_help_header	argv < 2 — startup banner second line
209	docprocessor_cli_path_invalid	empty / whitespace-only docs-directory arg
209	docprocessor_cli_error_resolving_path	filepath.Abs failure
209	docprocessor_cli_no_docs_found	loader returns 0 documents (was silent pre-209)
209	docprocessor_cli_format_summary	supported-formats line at run-start
209	docprocessor_cli_summary_header	summary section heading
209	docprocessor_cli_feature_line	per-feature line (verbose mode only)
209	docprocessor_cli_screen_line	per-screen line (verbose mode only)
209	docprocessor_cli_workflow_line	per-workflow line (verbose mode only)
209	docprocessor_cli_done	completion line with elapsed ms

All 18 IDs are present in pkg/i18n/bundles/active.en.yaml and asserted by TestRunCLI_BundleContainsAllRound209MsgIDs in cmd/docprocessor/main_test.go.

Key Interfaces

- loader.Loader — load documents from filesystem
- feature.FeatureMapBuilder — build feature maps from documents
- coverage.CoverageTracker — track feature verification status
- llm.LLMAgent — injected LLM for intelligent extraction (no hard dependency)
- i18n.Translator — externalised user-facing strings (CONST-046)

Configuration

Copy .env.example to .env and customise:

```
HELIX_DOCS_ROOT=./docs
HELIX_DOCS_AUTO_DISCOVER=true
HELIX_DOCS_FORMATS=md,yaml,html,adoc,rst
```

CONST-053 reminder. .env MUST be chmod 600 and is git-ignored. Only .env.example (placeholder values) is committed.

Testing

```
make test           # all tests
make test-race      # all tests with race detection
make test-cover     # all tests with coverage report
                    (coverage.html)
```

Test-type coverage (per CONST-050(B) 100%-test-type-coverage mandate):

Test type	File(s)	Real-in-fra?
Unit	pkg/**/*_test.go (excluding suffix-tagged), cmd/docprocessor/main_test.go	n/a (mocks OK)
Integration	pkg/loader/loader_integration_test.go	real fs
Stress	pkg/coverage/tracker_stress_test.go, pkg/docgraph/graph_stress_test.go,	real

Test type	File(s)	Real-in-fra?
Security	pkg/loader/loader_stress_test.go pkg/config/config_security_test.go, pkg/loader/loader_security_test.go, security_test.go	real
E2E	e2e_test.go	real binary
Automation	automation_test.go	real binary
Challenge	challenges/scripts/docprocessor_cli_challenge.sh (round 220) + 8 other Challenge scripts	real binary

Full per-test-file coverage matrix in [docs/test-coverage.md](#).

Challenge scripts

```
bash challenges/scripts/docprocessor_cli_challenge.sh      # round
                220 – CLI end-to-end + paired mutation
bash challenges/scripts/chaos_failure_injection_challenge.sh
bash challenges/scripts/ddos_health_flood_challenge.sh
bash challenges/scripts/host_no_auto_suspend_challenge.sh
bash challenges/scripts/no_suspend_calls_challenge.sh
bash challenges/scripts/scaling_horizontal_challenge.sh
bash challenges/scripts/stress_sustained_load_challenge.sh
bash challenges/scripts/ui_terminal_interaction_challenge.sh
bash challenges/scripts/ux_end_to_end_flow_challenge.sh
```

Every Challenge captures positive runtime evidence per CONST-035 / Article XI §11.9 and the §11.4 anti-bluff covenant. A passing Challenge is a claim that the feature works end-to-end for an end user, not merely that the binary exited 0.

Governance

- [CONSTITUTION.md](#) — module-specific tightenings on top of the canonical root.
- [CLAUDE.md](#) — AI-agent operating manual (cascaded from constitution submodule per CONST-049).
- [AGENTS.md](#) — generic agent manual peer of CLAUDE.md.
- Canonical root: [HelixConstitution](#).

Decoupling guarantee (CONST-051(B))

DocProcessor does NOT import any consuming-project namespace. The module ID is `digital.vasic.docprocessor`; no path under `pkg/**` or `cmd/**` references a specific consumer. Project-specific behaviour is injected at runtime via the Translator, LLMAgent, and Config contracts.

License

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